43 The archaeologist at the core of the Digital Workflow: AI, 3D architecture, and epistemological transparency for the reconstruction of Dur-Kurigalzu (Baghdad, Iraq)

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In the practice of virtual archaeology, the 3D reconstruction phase is frequently delegated to external IT technicians, fragmenting the interpretive chain and generating opaque, hard-to-verify "black box" models. This paper presents a hybrid workflow, developed and managed entirely by the archaeologist, for the philological reconstruction of the volumes and wall paintings of the Kassite palace of Dur-Kurigalzu (Iraq). The technical infrastructure integrates the 1940s Iraqi legacy data with primary photogrammetric data acquired during the University of Bologna's 2024-2026 campaigns. Within the workflow, the use of generative Artificial Intelligence was strictly confined to the digital restoration of the wall paintings. Conceived as a heuristic tool subordinated to human control, the AI was guided by strict prompts and geometric constraints to regenerate textures from decontextualized fragments and archival photos. Regardless of the specific software adopted for the pipeline, the architectural framework of the palace was built entirely on objective metric and spatial logic, ensuring total cross-platform replicability. Before becoming a visual output, the 3D modelling environment was utilized as a fully-fledged laboratory for virtual experimental archaeology. The model acted as an investigative tool to refute or validate historical hypotheses regarding the elevation: the modularity of the ancient spatial grid was tested by extruding and intersecting modern cubic primitives to verify their static and volumetric alignment with the excavation plans. Furthermore, to address the lack of internal supports in the massive reception halls, hypotheses for clerestory roofs were modelled. Their functional effectiveness was validated through a physically accurate lighting simulation, calculated based on the exact axial tilt and solar irradiance of the Late Bronze Age II in Mesopotamia. Finally, by migrating the model into a real-time game engine, the architecture becomes a spatial database. Here, the clear separation between verified excavation data and hypothetical volumetric interpolation is guaranteed by spatial proximity triggers: through interaction, users can look past the polygonal abstraction to retrieve the underlying real documentation on-screen (in situ photos, metadata, and plans). This project demonstrates how such an ecosystem empowers archaeologists to retain full authorial control, ensuring an interactive, falsifiable output that is structurally independent of any single software application.